Airport Management System – Progress Presentation

CSE 4304:
Software Project Lab - I

Team 5

Name : Sheikh Alimun Alahi (230042115)
Name : Rakibur Rahman Ananta (230042128)
Name: Irfan Sadik Chowdhury (230042147)

Project Overview: Simulating Airport Operations

Ourgoal is to simulatehowanairportactuallybehaveswhenresourcesarelimitedandtimeismoving,providingarealistic depiction of flight operations.

CLI-Based Simulation

A command-line interface system designed to simulate airport operations.

Flight Lifecycle Management

Focuses on the entire journey of flights, from scheduling to dep ar ture/ar r ival.

Resource Allocation

Manages gate and runway assignments under dynamic constraints.

File-Based System

Operates entirely through file I/O, without external databases or APIs.

System Architecture: Manager-Based Design

Oursystemis structured arounddistinctmanagers, each responsible fora specificfunctionalarea, promoting modularityand clear separationof concerns. This design emphasizes functional roles over specific code implementations.

Flight Management

Oversees all flight-related data and states.

Departure & Arrival

Handlesspecificlogicforoutgoingandincoming flights.

Resource Allocation

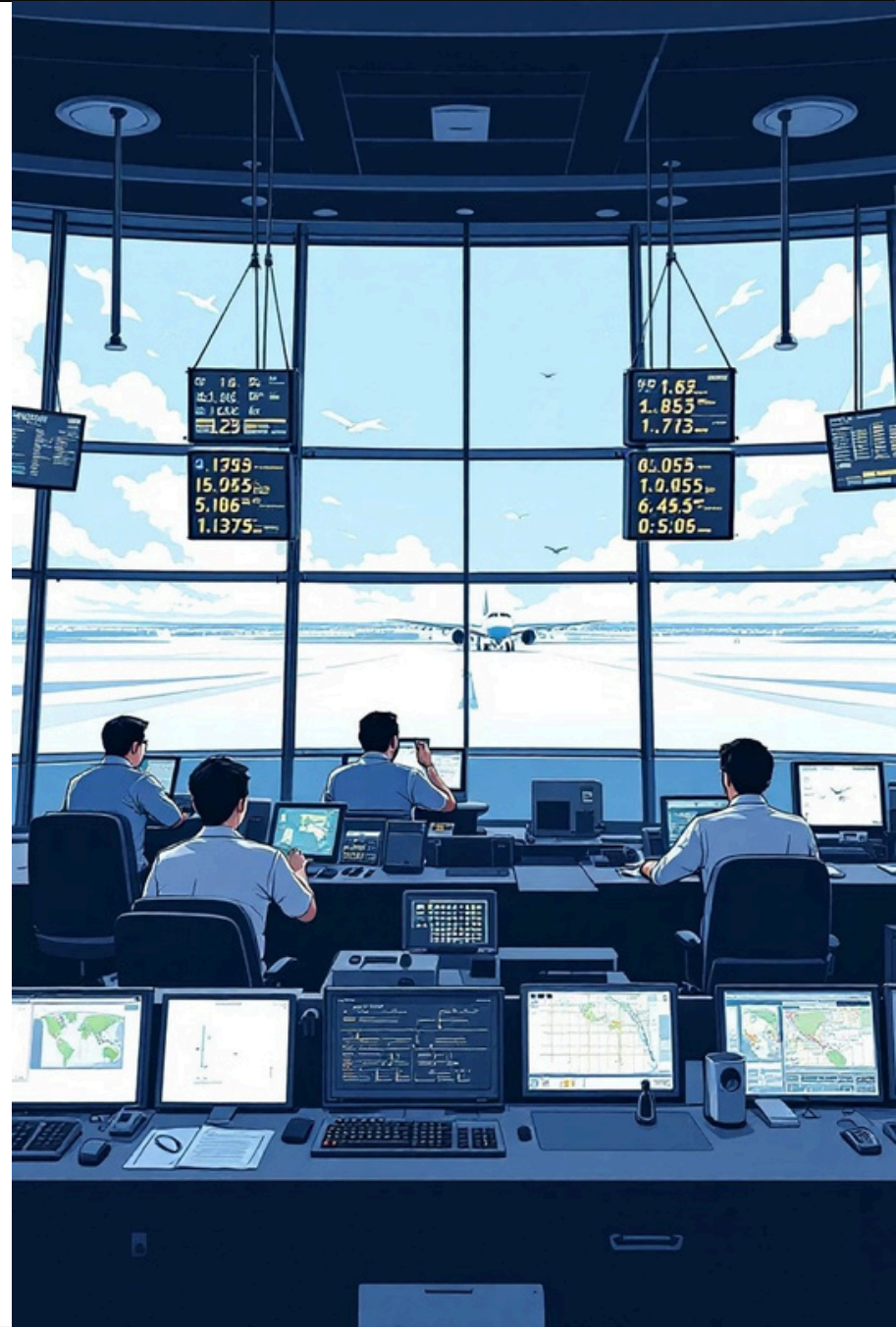
Manages gates and runways to prevent conflicts.

Passenger Management

Trackspassengerinformationandboarding status.

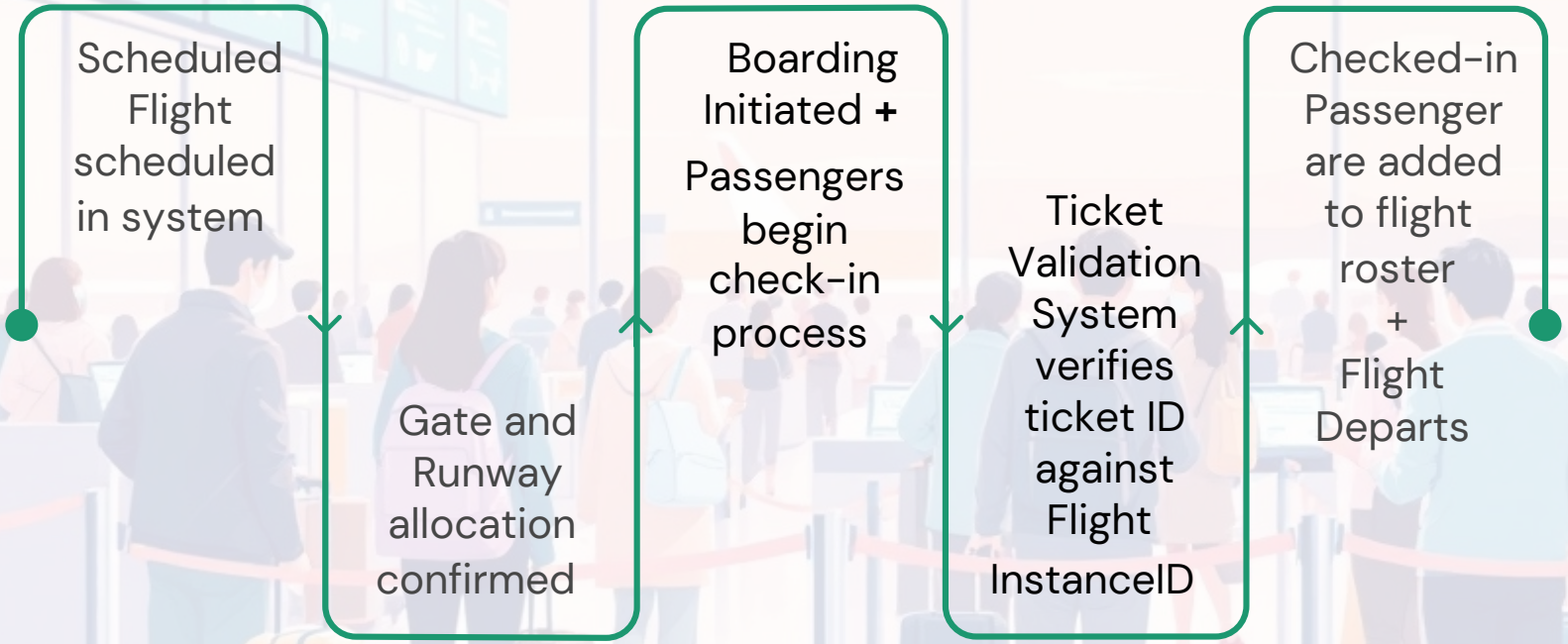
CLI Controller

The main interface for user interaction.



Passenger Check-In: Boarding State Logic

Duringboarding,passengerscheckinusingtheiruniqueticketIDs. Thesystemvalidateseach ticket against the flight's instanceID,ensuringaccuratepassenger-to-flightmatchingand preventingduplicatecheck-ins.



This validation process is critical to airport operations—it ensures data integrity, maintains accurate passenger manifests, and enables the system to track boarding progress in real time before departure.

Gantt Chart

Project Timeline

Task List	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11	WEEK 12	WEEK 13	WEEK 14
Planning and Models (Diagram&Modules)	Done													
Core Managers (Flight,Gate,Runway)			Done											
Mid Exam (Deployment Paused)						Paused								
Passenger &Booking								Partially Done						
Storage and Reports								Done						
Event System (Safety Logic and Weather)										To be Implemented				
Simulation and Chaos (Major Feature)												To be Implemented		
Final Sprint													To be Implemented	

*Current Status

Key Implemented Feature: Intelligent Resource Allocation

Acoreachievementofoursystem is the sophisticatedhandlingoflimitedresources,particularlygatesandrunways.This ensures efficient airport operations and prevents potential conflicts.

Limited Resources: Manages a finite pool of gates and runways.

Conflict Prevention: Ensures no two flights simultaneously use the same resource.

Occupancy Tracking: Continuously monitors occupied vs. free resource status.

Automatic Release: Resources are freed upon flight completion.

Passenger & File-Based Management: This approach allows us to concentrate on the intricacies of airport logic rather than database management.

A recent improvement includes automatic resource reallocation: if a gate is not immediately available, the system checks for completed flights to free up resources, minimizing delays. This showcases the intelligence built into our system.

Future Features: Roadmap to Completion

To finalizeourAirportManagementSystem,weareplanningseveraladvancedfeatures that will enhance realism, functionality, and user experience.

Chaos Mode

- Simulates real-world disruptions: flight delays, severe weather, technical issues
- Tests system resilience and recovery mechanisms

Simulation Engine

- Enables time-based airport simulation over extended periods
- Tracks system behavior and performance metrics

Role-Based Login

- Differentiated access for airport managers, staff, and administrators
- Secure authentication with role-specific per missions

Weather Updates

- Real-time weather integration affecting flight operations
- Dynamic impact on runway availability and scheduling

Booking Management

- Passenger booking and reservation system
- Integration with flight availability and seat allocation

These features will transform our system into a comprehensive, production-ready airport management platform.

Starting Demonstration.....